B.Sc. (Hons) in Software Development

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TRiM - A Multi Tenant SaaS Application

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Chapter 1

Introduction

1.1 Context

Small businesses such as barber shops that rely solely on walk-in customers face several operational challenges: limited business insights, customer friction when services are unavailable at their preferred time, and the burden of manual bookkeeping. Digitalising these operations bridges the gap between the business and its customers, leading to a more established market presence and streamlined day-to-day management.

TRiM is a platform designed to support the full range of barber shop operations, from customer bookings and payment processing to employee management and business analytics. Rather than serving a single business, *TRiM* is built to allow multiple businesses to operate on the platform independently, each with its own isolated data and user base. One instance of the software serves multiple businesses, this is known as multi-tenant architecture, and it is the core focus of this dissertation.

1.2 Project Objectives

The objective of this project is to deliver a real and robust software solution to local service-based businesses. The following goals have been defined:

- **Customer Booking System**: Enable customers to book services online, select their preferred barber, choose from categorized services, and pick available time slots while the system prevents double bookings and scheduling conflicts.
- **Payment Integration**: Integrate Stripe for secure online payment processing, supporting both full online payments and deposit-based bookings where

customers pay a percentage upfront and settle the outstanding balance in-shop, driven by real client requirements.

- **Business Administration**: Provide business owners with a comprehensive admin dashboard to manage service categories, individual services with pricing and duration, and employee (barber) profiles. Include analytics such as total revenue, booking statistics, and popular services.
- **Barber Management**: Allow barbers to manage their own availability by setting weekly working hours, defining break periods, and viewing/managing their upcoming bookings with the ability to confirm, complete, or handle cancellations and no-shows.
- **Customer Management**: Enable administrators to view customer history, track no-show counts, and blacklist problematic customers to protect business operations.
- **Multi-Tenant Architecture**: Design the platform to support multiple independent businesses, each with their own subdomain (business slug), isolated data, and separate admin/barber/customer user bases.
- **Role-Based Access Control**: Implement distinct user roles (CUSTOMER, BARBER, ADMIN) with appropriate permissions and tailored interfaces for each role's specific needs.

1.3 Dissertation Focus

This dissertation focuses primarily on the architectural decisions behind *TRiM*, particularly the database design patterns and data isolation techniques employed. Through *TRiM*, we explore how to build a multi-tenant software application that is both scalable and robust, leveraging efficient design patterns that allow the system to grow without significant cost or performance overhead. We also investigate how multiple businesses can share the same platform whilst maintaining strong performance guarantees and strict security boundaries.

1.4 Success Metrics

From a functional standpoint, the system must handle all core operations reliably: bookings should complete without conflicts, payments should process securely, and the admin dashboard should provide accurate insights into business operations.

Beyond functionality, performance represents a critical area of evaluation. The system will be benchmarked under various conditions to assess its resilience. This includes measuring response times for key API endpoints, evaluating database performance as the number of tenants and bookings increases, and verifying that heavy usage by one business does not degrade the experience for others. Tenant isolation extends beyond security considerations, it also ensures fair distribution of system resources across all users.

Security constitutes another essential metric. With multiple businesses and their customer data at stake, the system must demonstrate robust protection mechanisms. Cross-tenant data leakage would represent a critical failure, necessitating thorough testing of all isolation mechanisms to prevent such occurrences.

Finally, usability remains a key consideration. The primary objective of *TRiM* is to streamline operations for business owners and their customers. An unintuitive interface or cumbersome booking flow would undermine the value of the underlying architecture.

1.5 Chapters Overview

Chapter 2 – Methodology

Outlines the development approach adopted for this project, including the agile techniques and project management tools used throughout the development lifecycle.

Chapter 3 – Technology Review

Reviews multi-tenant database architectures and the key technologies underpinning the platform, providing the theoretical foundation for the design decisions made.

Chapter 4 – System Design

Presents the overall system architecture, REST API design, database schema, and the multi-tenant data isolation strategy.

Chapter 5 – System Evaluation

Evaluates the system through performance benchmarking, load testing, and security analysis to assess how the platform behaves under realistic conditions.

Chapter 6 – Conclusion

Summarises the key findings, reflects on the project objectives, and discusses potential future work and improvements.

Chapter 2

Methodology

2.1 Development Approach

2.1.1 Agile Development Techniques Applied

- Kanban Board with GitHub Projects - Sprints per user story

2.1.2 Version Control (GitHub)

- InteliJ IDE + VS Code + Codespaces

2.1.3 Requirements Gathering

- Real client - User stories

2.2 Generative AI Usage

2.2.1 Code Generation with GitHub Copilot

2.2.2 Testing Automation with Claude Code

2.2.3 Code Review

2.3 Research Approach

2.3.1 Literature Search

- Search strings and databases used - Inclusion and exclusion criteria

2.3.2 Choosing the Technology Stack

- Reasons for selecting Java, Spring Boot, React, PostgreSQL, etc.

2.4 Evaluation Methodology

2.4.1 Benchmarking With Gaitling Load Testing

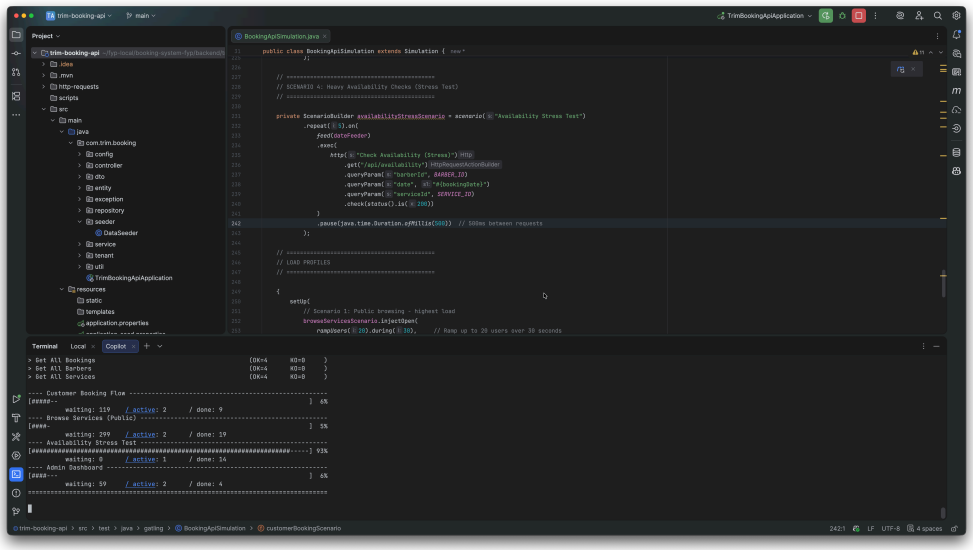

Figure 2.1: Benchmarking API in IntelliJ IDE.

2.4.2 Test Data Generation With java-seeder

2.4.3 Performance Metrics

Chapter 3

Technology Review

3.1 Multi-Tenant Database Architectures

A multi-tenant system provides one shared instance of an application and its infrastructure to multiple users, referred to as tenants, while maintaining the logical separation of their data. In a SaaS context, tenants are typically organisations or businesses that each require their own private view of the data. How that separation is achieved at the database level is one of the most important design decisions in a SaaS application, as it determines the system's performance, security, operational cost, and ability to scale [1].

3.1.1 Different Types of Multi-Tenant Database Designs

The multi-tenant database design can be broken down into three primary isolation models, as defined by Karatąs et al. [2] and illustrated in Figure 3.1.

Separate Application, Separate Database gives each tenant a fully independent application instance and database. Data isolation is guaranteed: there is no shared infrastructure at the database layer, so there is no risk of one tenant accessing another's data. The trade-off is cost and operational overhead. Every new tenant requires provisioning an additional stack, and hardware resources cannot be shared across tenants, making this model expensive to operate at scale [2].

Shared Application, Separate Database retains a single application instance but still provides each tenant with its own database. This reduces application-layer overhead while preserving data isolation. It's still expensive in terms of storage and database management, since the number of databases grows linearly with the number of tenants.

Shared Application, Shared Database is the most resource-efficient model. All tenants share the same application and the same database, with data partitioned logically rather than physically. This model has two variants. In *Shared*

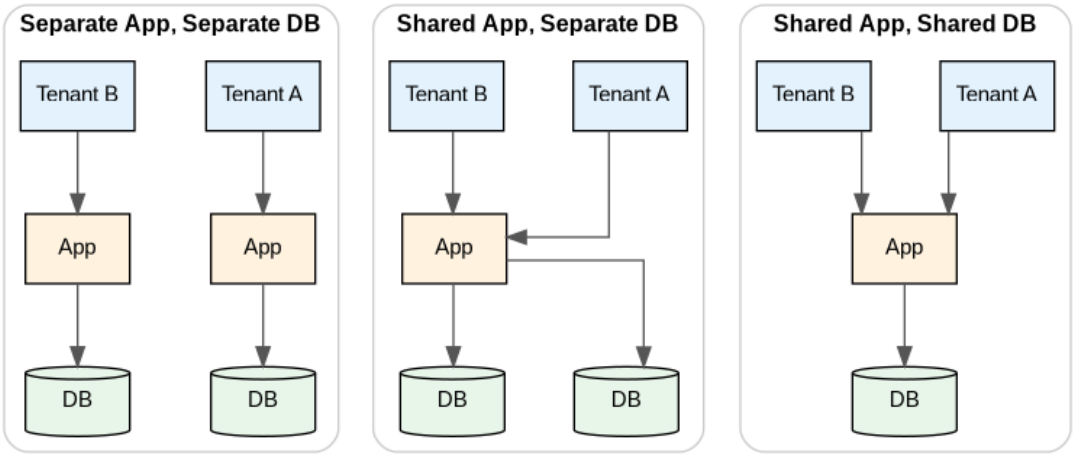

Figure 3.1: Multi-tenant database isolation patterns

Database, Separate Schema, each tenant receives its own set of tables within the shared database. In *Shared Database, Shared Schema*, all tenants share the same tables, and their rows are distinguished by a `tenant_id` column. The shared schema variant offers the highest resource utilisation and the lowest per-tenant cost, but it places a burden on the data isolation mechanisms, since a missing or incorrect filter predicate could expose one tenant's data to another [1].

3.1.2 Trade-offs: Isolation vs. Efficiency vs. Cost

The selection of an isolation model is not a decision that can optimise all dimensions simultaneously. Wang et al. [1] established the foundational tension: higher isolation means better security but limits the number of tenants a system can support on fixed hardware. Zhou et al. [3] quantify this directly on IBM DB2, demonstrating that a shared table model (S3) supports approximately 400 concurrent tenants on the same hardware that a separate database model (S1) can sustain only around 8. The capacity advantage of the shared approach is significant, but it depends on effective isolation being built into the system.

Cost is yet another concern. Wang et al. [1] categorise multi-tenant system costs across three dimensions: infrastructure (hardware, software, and utilisation), management (lifecycle, monitoring, and backup), and development (tenant-specific customisation). Shared architectures reduce infrastructure and management costs substantially. Aulbach et al. [4] and Yaish and Goyal [5] both demonstrate that shared schemas lower Total Cost of Ownership by eliminating per-tenant schema provisioning and by enabling self-service configuration. However, neither work

quantifies how the cost of securing a shared schema compares to the cost of stronger physical isolation.

Performance also needs to be taken into account. Wang et al. [1] measured that adding Label-Based Access Control (LBAC) - a database-level security mechanism - reduced throughput by 15% compared to less secure approaches. Ni et al. [6] propose the ADAPT methodology, which improves query performance by 2–3× over static shared-schema approaches by dynamically clustering attributes based on workload, but their evaluation explicitly assumes application-level filtering and tests no security mechanisms. The pattern is consistent across the literature: studies that optimise performance assume the security problem is solved elsewhere, and studies that address security do not measure the performance impact of their mechanisms [2, 7].

3.1.3 Data Isolation Techniques

Within the shared database, shared schema model, the mechanisms used to enforce tenant data isolation fall into two categories: filter-based and permission-based.

Filter-based isolation is implemented at the application layer by appending a tenant identifier predicate to every query, typically expressed as `WHERE tenant_id = ?`. Wang et al. [1] identify this as the most common approach in practice due to its simplicity. Its critical weakness is that isolation is only as reliable as the application code: a missing predicate, a buggy query, or a SQL injection vulnerability can expose data across tenant boundaries. Gupta, Kumar, and Agnihotri [8] extend this approach with session-based identifiers that rotate periodically, reducing injection risk, but their mechanism remains dependent on the application being free of errors.

Permission-based isolation enforces tenant boundaries at the database engine level, independently of the application. PostgreSQL Row-Level Security (which has been used in this project) is a prominent example: policies are attached directly to tables, and the database engine automatically appends the isolation predicate to every qualifying query before execution, regardless of what the application sends. Factor et al. [9] formalise the principles underpinning this approach in their SLIM (Secure Logical Isolation for Multi-tenancy) model, based on least privilege, tenant containment, escalation avoidance, and minimising attack surface. The advantage of database-level isolation is that it acts as a safety net: even if the application layer has a bug, the database will not return rows belonging to another tenant. The documented trade-off is a performance overhead from the additional policy evaluation on every query.

3.2 Related Works

3.2.1 Proposed Multi-Tenant SaaS Database Design Patterns

Several studies have proposed specific techniques for managing multi-tenant data that are directly relevant to the design decisions made in this project.

Ni et al. [6] introduce **ADAPT** (Adaptive Database Schema Design for Multi-Tenant Data Management), a methodology that dynamically reorganises table schemas by clustering attributes based on observed query workloads. ADAPT achieves a 2–3× throughput improvement over static shared-schema designs and demonstrates reasonable scalability up to 4,000 tenants, at which point the alternative individual-table approach degrades severely. Its limitation for this project's context is that it assumes application-level filtering and does not evaluate how it would work with database-level security policies.

Mangwani and Tokekar [10] evaluate microservice architectures as a scalability mechanism for multi-tenant SaaS systems. Their findings show that at low concurrency (around 50 users), monolithic and microservice architectures perform similarly, but at high load (300+ users), microservices offer lower response times, higher throughput, and approximately 11% better CPU utilisation. This supports the case for modular application design but treats the database as a black box, with no evaluation of how schema design choices interact with service decomposition.

Sharma and Kaur [11] demonstrate MongoDB sharding as an alternative to relational multi-tenant designs, enabling horizontal scaling through data partitioning across nodes with automatic load balancing. While this illustrates the growing relevance of NoSQL approaches, the study does not compare its performance or isolation guarantees against optimised relational alternatives, leaving the trade-off between schema flexibility and data isolation unknown.

Most recently, Yin et al. [12] propose the MTD-DS benchmark, arguing that existing benchmarks such as TPC-H and TPC-C fail to capture SLA-relevant metrics like query completion time per tenant and tenant prioritisation. Their work introduces SLA-aware pricing models including a Query-Level SLA-Aware (QLSA) model that ties provider cost to actual service level objective satisfaction. While this framework represents a conceptual advance, it remains a benchmark model with no empirical validation against real-world systems.

3.2.2 Literature Reviews

Two prior systematic reviews are most directly relevant to this work and establish the context into which this project's design decisions fit.

Karataş et al. [2] conducted a broad systematic mapping study covering 407 papers from 2000 to 2017. Their key quantitative finding is that 65% of papers preferred a shared database strategy, and 58% were ideological proposals without implementation. The study is descriptive rather than evaluative: it reports which strategies are preferred but does not analyse why or evaluate combined approaches. Its 2017 cutoff also means it predates significant developments in containerised deployment and modern database security tooling.

Battula [7] provides a more recent (2024) systematic review focused specifically on security mechanisms in multi-tenant cloud databases, analysing 25 papers from 2012 to 2024. It comprehensively covers encryption, access control, and threat detection, but treats performance and scalability as secondary concerns. Like Karataş et al., it evaluates mechanisms in isolation and identifies no integrated approaches combining database-level security with performance optimisation techniques.

The gap identified by both reviews, and confirmed by the broader set of papers analysed, is the absence of empirical evaluation of combined approaches. No study evaluates how a database-level security mechanism such as RLS performs when applied alongside a shared schema optimised for multi-tenant query patterns, nor how much of the performance benefit claimed by approaches like ADAPT survives the addition of policy-based isolation. Zhu et al. [13] highlight an additional dimension of this problem: optimising database performance for one tenant can degrade it for others, a conflict that existing research leaves unresolved.

This project addresses a subset of this integration gap directly. TRiM adopts a Shared Application, Shared Database, Shared Schema design, the pattern associated with the highest resource efficiency, and enforces tenant isolation using PostgreSQL Row-Level Security rather than application-layer filtering alone. The load testing results presented in Chapter 5 provide empirical evidence on the performance overhead of this combination under real data volumes, contributing practical findings to the field.

Chapter 4

System Design

4.1 System Overview

4.1.1 Architecture Diagram

4.1.2 RESTful API Design

4.1.3 Technology Stack

4.2 Multi-Tenant Database Design

4.2.1 Entity-Relationship Model

4.2.2 Migrating From Single to Multi-Tenant

4.2.3 Tenant-Aware Schema

4.2.4 Row Level Security vs Application-Level Isolation

4.3 Multi-Tenant Implementation

4.3.1 TenantFilter & TenantContext

4.3.2 Subdomain-based Routing

4.3.3 Cross-Subdomain Token Exchange

4.4 Business Operations

4.4.1 Booking & Payment Flow

4.4.2 Race Condition Prevention When Booking

4.4.3 Authentication & Security

4.4.4 JWT & Role-Based Access Control

4.5 Frontend Architecture

4.5.1 Component Structure

and algorithms to illustrate your design.

Chapter 5

System Evaluation

5.1 Performance Evaluation

5.1.1 Database Gaitling Benchmarking - Load Testing

5.2 Security Evaluation

5.2.1 Tenant Isolation Verification

5.3 Critical Analysis

5.3.1 Strengths

5.3.2 Limitations

Chapter 6

Conclusion

6.1 Future Work

Briefly summarise your context and objectives. Remind the reader about the overall rationale and goals of the project. Highlight your findings from the System Evaluation chapter.

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